(4) The chances of A, B and C becoming manager of a certain company are 5 : 3 : 2. The probabilities that the office canteen will be improved if A, B, and C become managers are 0.4, 0.5 and 0.3 respectively. If the office canteen has been improved, what is the probability that B was appointed as the manager?

(5) An advertising executive is studying television viewing habits of married men and women during prime time hours. Based on the past viewing records he has determined that during prime time wives are watching television 60% of the time. It has also been determined that when the wife is watching television, 40% of the time the husband is also watching. When the wife is not watching the television, 30% of the time the husband is watching the television. Find the probability that (i) the husband is watching the television during the prime time of television (ii) if the husband is watching the television, the wife is also watching the television.

EXERCISE 12.5

Choose the correct or most suitable answer from the given four alternatives

(1) Four persons are selected at random from a group of 3 men, 2 women and 4 children. The probability that exactly two of them are children is

(1) $\frac{3}{4}$ (2) $\frac{10}{23}$ (3) $\frac{1}{2}$ (4) $\frac{10}{21}$

(2) A number is selected from the set $\{1,2,3,...,20\}$. The probability that the selected number is divisible by 3 or 4 is

(1) $\frac{2}{5}$ (2) $\frac{1}{8}$ (3) $\frac{1}{2}$ (4) $\frac{2}{3}$

(3) A, B, and C try to hit a target simultaneously but independently. Their respective probabilities of hitting the target are $\frac{3}{4}, \frac{1}{2}, \frac{5}{8}$. The probability that the target is hit by A or B but not by C is

(1) $\frac{21}{64}$ (2) $\frac{7}{32}$ (3) $\frac{9}{64}$ (4) $\frac{7}{8}$

(4) If A and B are any two events, then the probability that exactly one of them occur is

(1) $P(A\cup\overline{B})+P(\overline{A}\cup B)$ (2) $P(A\cap\overline{B})+P(\overline{A}\cap B)$

(3) $P(A)+P(B)-P(A\cap B)$ (4) $P(A)+P(B)+2P(A\cap B)$

(5) Let A and B be two events such that $P\left(\overline{A\cup B}\right)=\frac{1}{6}$, $P\left(A\cap B\right)=\frac{1}{4}$ and $P\left(\overline{A}\right)=\frac{1}{4}$. Then the events A and B are

(1) Equally likely but not independent (2) Independent but not equally likely

(3) Independent and equally likely (4) Mutually inclusive and dependent

(6) Two items are chosen from a lot containing twelve items of which four are defective, then the probability that at least one of the item is defective

(1) $\frac{19}{33}$ (2) $\frac{17}{33}$ (3) $\frac{23}{33}$ (4) $\frac{13}{33}$

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